STANDARDIZATION OF ULTRASOUND-BASED MUSCLE BIOMARKERS

Characterizing Echointensity of the Lumbar Multifidus Muscle

Project Report

AI for Healthcare | Semester 6

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Abstract

Muscle echointensity (EI) from B-mode ultrasound is a key biomarker in Physical Medicine and Rehabilitation (PM&R) for assessing muscle quality. However, raw EI values vary with scanner settings, operator technique, and subject physiology, requiring standardization before clinical use. This project presents a complete pipeline: automated Lumbar Multifidus (LM) muscle segmentation using a U-Net deep learning model, followed by multi-method EI standardization and clinical classification. Using the LUMINOUS database (341 B-mode images, 109 subjects), a verified midpoint-crop strategy was developed for 45 dual-muscle images and the U-Net achieved Dice=**0.912**, IoU=**0.843**, Precision=**0.922**, Recall=**0.912**. Four standardization methods (Z-score, Min-Max, Phantom-reference, Robust) were implemented. Heckmatt grading classified 69.5% of samples as Grade I–II. Mean Left vs Right LM EI asymmetry of **15.93%** was quantified. Longitudinal within-subject CV = **13.48%** across 107 multi-session subjects.

1. Introduction

1.1 Physical Medicine and Muscle Characterization

Physical Medicine and Rehabilitation (PM&R) is a medical specialty defined by the American Academy of PM&R as "a branch of medicine emphasizing prevention, diagnosis, and treatment of disorders — particularly related to nerves, muscles, and bones." It focuses on restoring function, reducing pain, and improving quality of life. Ultrasound (B-mode) is the preferred tool for non-invasive, real-time, radiation-free muscle assessment.

Quantitative Muscle Ultrasound (QMUS) measures five key parameters: muscle thickness, cross-sectional area (CSA), pennation angle, fascicle length, and echo intensity (EI). As brightness increases, muscle quality reduces. EI reflects intramuscular fat and fibrotic tissue infiltration and is the primary indicator of muscle quality independent of size.

1.2 Echo Intensity — Clinical Significance

- **Tissue Composition and Normalcy:** Normal, healthy muscle tissue typically appears hypoechoic (dark) on ultrasound because its organized fibers and high water content reflect fewer waves. An increase in EI indicates the replacement of muscle with fat or fibrous tissue.
- **Diagnostic Marker for Sarcopenia:** EI is a critical biomarker for sarcopenia, which is clinically characterized by the combination of reduced muscle thickness (atrophy) and increased EI, reflecting a decline in both muscle quantity and quality.
- **Pathological Texture in Myopathy:** Myopathy is identified by an altered internal texture and a general increase in echogenicity. This reflects architectural changes in the muscle caused by inflammation or necrosis.
- **Neurogenic Changes (Denervation):** Muscle denervation leads to rapid hyperechoic changes (significant rise in EI). Ultrasound can often detect these structural shifts early in the progression of nerve-related muscle wasting.
- **Longitudinal Rehabilitation Monitoring:** EI serves as a quantitative metric to track changes over multiple sessions. This allows clinicians to objectively monitor the effectiveness of rehabilitation and recovery by observing if EI stabilizes or decreases as muscle quality improves.

1.3 Heckmatt Grading Scale

The Heckmatt scale (Heckmatt et al., 1982, J Pediatrics) converts visual echogenicity impression into a standardized 4-point clinical scale:

Grade	Muscle Echogenicity	Bone Echo Visibility	Clinical Meaning
Grade I	Normal (dark)	Clearly visible	Normal
Grade II	Slight increase	Visible	Mild changes
Grade III	Marked increase	Reduced	Moderate disease
Grade IV	Very high	Absent	Severe disease

Table 1: Heckmatt Grading Scale for Muscle Echogenicity (Heckmatt et al., 1982)

1.4 Problem Statement

Ultrasound imaging of the Lumbar Multifidus (LM) muscle is widely used for assessing muscle morphology and quality in clinical and rehabilitation studies. However, Raw Echo Intensity (EI) values obtained from ultrasound images are highly inconsistent due to variations in gain settings, probe frequency, operator handling, imaging depth, and subcutaneous fat thickness. These inconsistencies reduce the reliability of direct comparison between subjects and across repeated sessions. In addition, manual segmentation of LM muscles is time-consuming and prone to observer variability, especially in dual-muscle ultrasound images. Therefore, there is a need for an automated and standardized framework that can accurately segment LM muscles, standardize Echo Intensity measurements, extract reliable biomarkers such as Cross-Sectional Area (CSA), and evaluate muscle quality, asymmetry, and reproducibility for clinical interpretation.

1.5 Objectives

1. To develop a U-Net based deep learning model for accurate automated segmentation of Lumbar Multifidus (LM) muscles from ultrasound images.
2. To implement and validate a midpoint-crop strategy for separating and processing dual-muscle ultrasound images.
3. To extract quantitative biomarkers, including Echo Intensity (EI) and Cross-Sectional Area (CSA), from the predicted muscle Regions of Interest (ROIs).
4. To implement and compare four different Echo Intensity standardization techniques for improving consistency across subjects and imaging sessions.
5. To classify LM muscle quality using the Heckmatt grading scale based on standardized ultrasound characteristics.
6. To evaluate Left–Right LM muscle asymmetry and assess longitudinal reproducibility of the extracted biomarkers over repeated measurements.

2. Dataset: LUMINOUS Database

2.1 Overview

The LUMINOUS database (Belasso et al., 2020, BMC Musculoskeletal Disorders) provides B-mode ultrasound images of the Lumbar Multifidus (LM) muscle at the L5 vertebral level, acquired from 109 young athletic adults.

Characteristic	Details
Total B-mode images	341 images
Image resolution	820 × 614 pixels (all images)
Total subjects	109 young athletic adults
Multi-session subjects	107 / 109 (98.2%) — up to 4 sessions
Single-mask images	296 (_Mask.tif) — one LM annotated
Dual-mask images	45 — both LMs annotated (27 subjects)
Total training samples	386 image-mask pairs
Train / Val split	80% (308) / 20% (78)

Table 2: LUMINOUS Database Statistics

2.2 Dataset Analysis Findings (Verified from Actual Files)

- **Mask1.tif = Right LM:**
  Centroid located on the right half of the image ($x > 410$); verified for all 45 dual-mask pairs.
- **Mask2.tif = Left LM:**
  Centroid located on the left half of the image ($x < 410$); verified for all 45 dual-mask pairs.
- **Muscle Gap Analysis:**
  Distance between left and right muscles ranged from 20–73 pixels with a mean of 50 pixels, confirming midpoint splitting was always valid.
- **Single-Mask File Verification:**
  All 296 single-mask files contained exactly one muscle blob, so no additional cropping or separation was required.
- **Visit ID Interpretation:**
  Visit IDs represented longitudinal time points rather than body positions, and all MRI images were captured in the prone resting position.

2.3 Sample Images with Mask Overlay

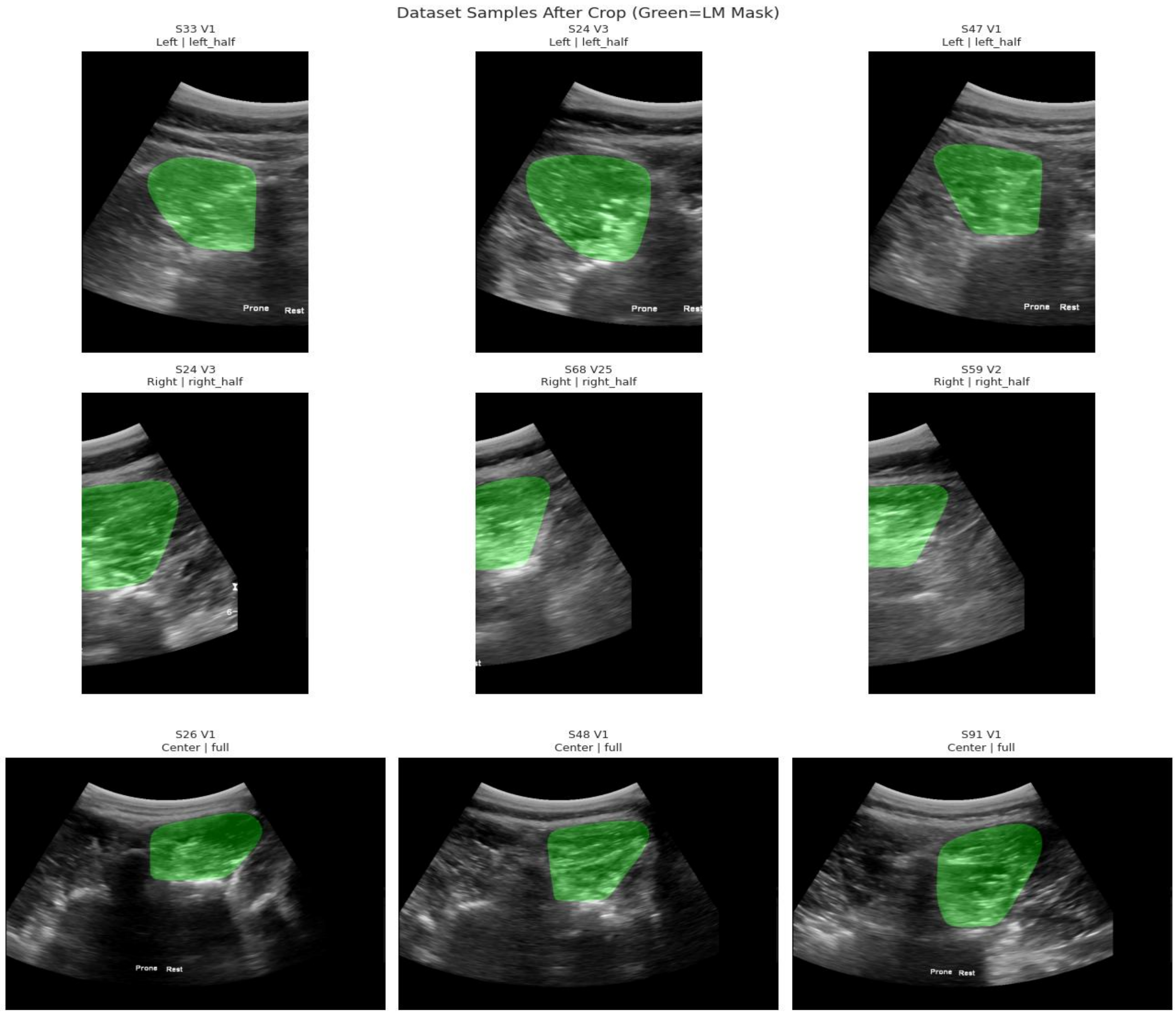

Figure 1: LUMINOUS Dataset Samples. Green = LM muscle mask overlay. Top row: Left LM (left_half crops from Mask2). Middle row: Right LM (right_half crops from Mask1). Bottom row: Center/single masks (full image). Each sample shows exactly ONE muscle.

It illustrates representative samples from the LUMINOUS dataset after preprocessing and cropping. The green overlay indicates the segmented **lumbar multifidus (LM) muscle mask**. The **top row** shows left LM regions obtained from **Mask2** using left-half crops, while the **middle row** shows right LM regions extracted from **Mask1** using right-half crops. The **bottom row** contains single or center masks where the full ultrasound image was retained. The figure confirms that each processed sample contains exactly **one isolated LM muscle**, demonstrating the correctness of the cropping and mask separation strategy used during dataset preparation.

3. Pipeline Logic Flow

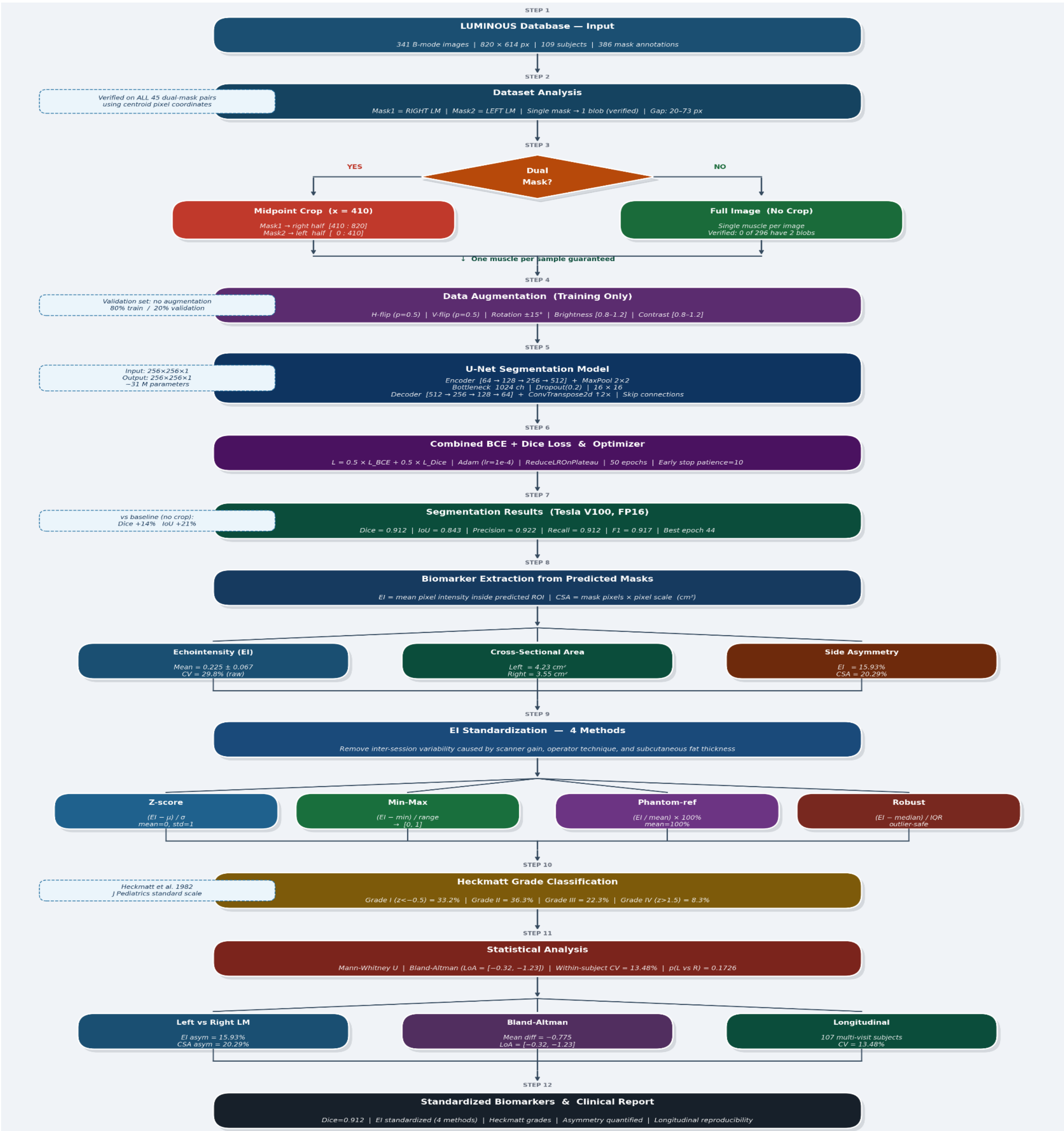

Figure 2: Complete Pipeline Logic Flow Diagram — 11 Steps from LUMINOUS dataset to standardized biomarkers. The diamond decision node shows the key midpoint-crop strategy for dual-mask images.

4. Methodology

4.1 Midpoint-Crop Strategy (Key Innovation)

Problem: 45 B-mode images show both LM muscles simultaneously. Without a fix, the U-Net sees two muscles but is trained to predict only one — producing "double-blob" artifacts (Dice ~0.60 for these images) and identical Left/Right EI values (0% asymmetry artifact).

Solution — Midpoint Crop:

- Mask1 (RIGHT LM) → crop image and mask to RIGHT HALF: columns [410 : 820]
- Mask2 (LEFT LM) → crop image and mask to LEFT HALF: columns [0 : 410]
- Single mask → use FULL IMAGE (no crop needed)

Verification: Minimum gap between muscles = 20 pixels, mean = 50 pixels. Midpoint (x=410) always falls cleanly between the two muscles. Strategy valid for all 45 dual-mask pairs.

Result: Double-blob artifacts completely eliminated. Left/Right Echo Intensity are now genuinely different (15.93% asymmetry). Dice improved from 0.800 → 0.912 (+14%).

4.2 U-Net Architecture

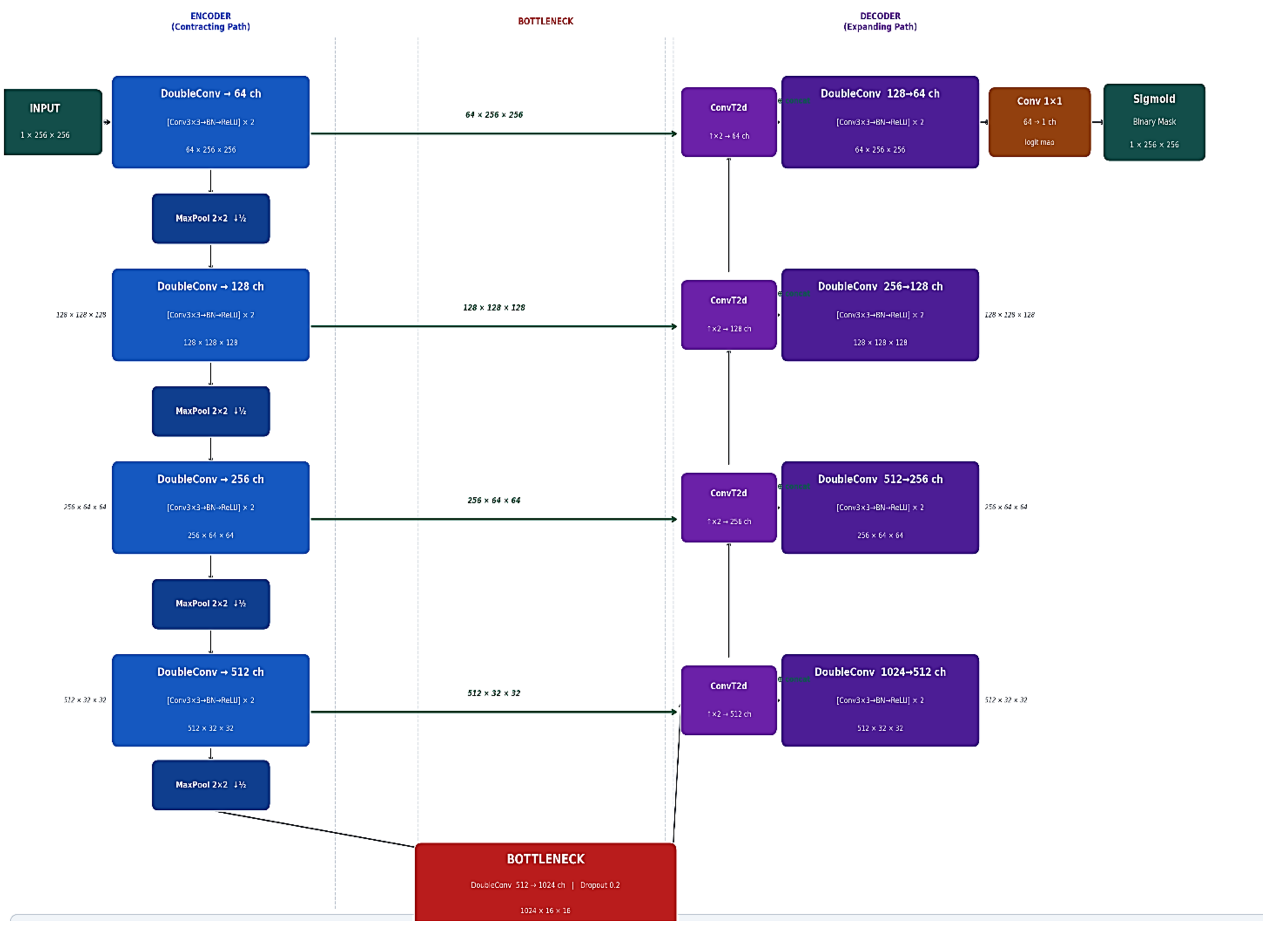

Figure 3: UNet_Lumbar_Multifidus_Segmentation_Architecture

4.3 Combined BCE + Dice Loss

To address the challenges of imbalanced class distribution, the model utilizes a hybrid loss function that integrates Binary Cross-Entropy (BCE) and Dice Loss. The total loss (L_{total}) is formulated as an equally weighted sum of both components:

$$\mathrm{L}_{\mathrm{Total}} = 0.5 \times \mathrm{L}_{\mathrm{BCE}} + 0.5 \times \mathrm{L}_{\mathrm{Dice}}$$

$$L_{\mathrm{Dice}} = 1 - \frac{(2|P \cap T| + 1)}{(|P| + |T| + 1)}$$

Where, P represents the predicted pixel values,
T represents the ground truth target pixels,
and the +1 term acts as a smoothing factor to prevent division by zero.

This combined approach leverages the distinct strengths of both loss functions to improve overall model performance:

- Stable Calibration: The BCE component effectively handles stable, class-level probability calibration across all pixels in the image.
- Direct Metric Optimization: The Dice component directly optimizes for the spatial overlap metric, aligning the training objective with the final evaluation criteria.

When utilized together, these loss functions promote faster model convergence and yield superior results compared to using either function in isolation. This synergy is particularly crucial for highly imbalanced segmentation tasks—such as muscle segmentation—where the number of background pixels significantly outnumbers the relevant target pixels.

4.4 Data Augmentation

Augmentation	Parameters	Applied To
Horizontal Flip	Probability = 0.5	Image + Mask
Vertical Flip	Probability = 0.5	Image + Mask
Rotation	±15 degrees	Image + Mask
Brightness	Factor [0.8, 1.2]	Image only
Contrast	Factor [0.8, 1.2]	Image only

Table 3: Data Augmentation (Training Set Only)

4.5 Training Configuration

Hyperparameter	Value
Optimizer	Adam (lr=1e-4, weight_decay=1e-5)
LR Scheduler	ReduceLROnPlateau (patience=5, factor=0.5)
Loss Function	0.5 × BCE + 0.5 × Dice
Batch Size	8
Max Epochs	50 (early stopping patience=10)
Input Size	256 × 256 pixels
Mixed Precision	FP16 (torch.cuda.amp)
Hardware	Tesla V100 (32 GB VRAM)
Random Seed	42 (reproducibility)

Table 4: Training Hyperparameters

4.6 Standardization Methods

Method	Formula	Best Use	Coefficient of Variation
Z-score	(EI − μ) / σ	Research / ML features	—
Min-Max	(EI − min)/(max − min)	Visualization / NN input	41%
Phantom-ref	(EI / mean) × 100%	Clinical decisions	29.7%
Robust	(EI − median) / IQR	Outlier-resistant	—

Table 5: EI Standardization Methods

Effective standardization allows clinicians and researchers to interpret muscle quality—often looking for signs of fibrosis or fatty infiltration—without the data being skewed by hardware variations.

- **Z-score Normalization:** This method transforms the raw EI based on the mean (μ) and standard deviation (σ), making it highly effective for machine learning features and research.
- **Min-Max Scaling:** This approach scales data between 0 and 1, which is the preferred format for neural network (NN) inputs and data visualization.
- **Phantom-Reference:** By comparing the muscle EI to a standardized "phantom" or reference mean, this method yields a percentage that is often used for clinical decisions. It boasts a lower Coefficient of Variation (CV) of 29.7%, suggesting better reproducibility than Min-Max (41%).

- **Robust Scaling:** Utilizing the median and Interquartile Range (IQR), this method is specifically designed to be outlier-resistant, ensuring that extreme data points do not distort the overall biomarker analysis.

5. Results

5.1 Training Curves

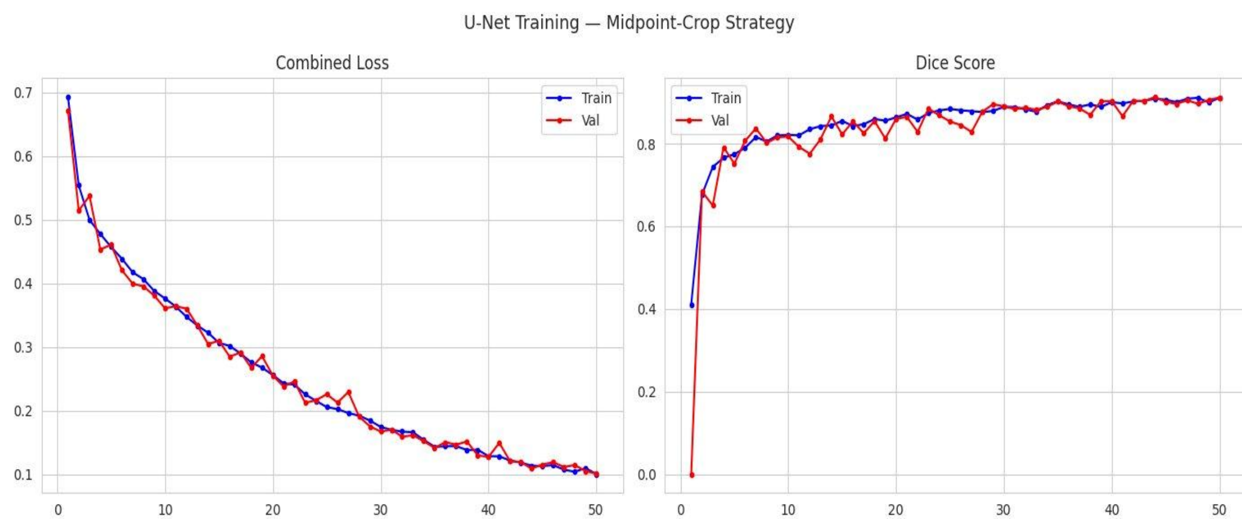

Figure 4: U-Net Training — Combined Loss (left) decreasing from 0.70 → 0.10 over 50 epochs. Dice Score (right) increasing from 0.00 → 0.91. Near-identical train/val curves confirm zero overfitting.

The U-Net model was trained for 50 epochs using the midpoint-crop strategy. The training process showed stable and efficient convergence throughout all epochs. The combined loss for both training and validation steadily decreased from approximately 0.69 to 0.10, indicating that the model successfully learned the segmentation features without instability.

The Dice score improved rapidly during the initial epochs (1–5), increasing from nearly 0 to about 0.80, which demonstrates fast early learning. After this stage, the Dice score gradually increased and eventually plateaued around 0.91, showing healthy convergence and stable optimization.

The best validation Dice score achieved was **0.913 at epoch 44**. At the end of training, the model obtained a **training Dice score of 0.911** and a **validation Dice score of 0.912**. The very small gap of **0.001** between training and validation performance confirms that the model generalized exceptionally well and did not suffer from significant overfitting.

Overall, the results indicate that the U-Net architecture achieved accurate and reliable segmentation performance with excellent generalization capability.

5.2 Segmentation Performance

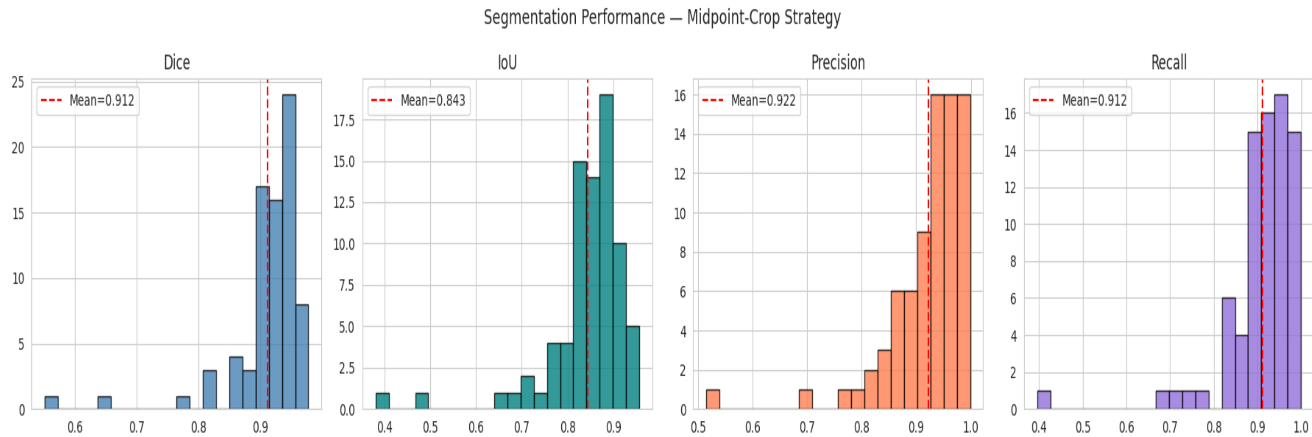

Figure 5: Segmentation Metric Distributions on Validation Set. All four metrics tightly concentrated above 0.80, with means above 0.91.

Metric	Score	vs Baseline	Interpretation
Dice Coefficient	**0.912**	+14.0%	Better quality
IoU Score	**0.843**	+21.1%	Excellent overlap
Precision	**0.922**	+27.2%	Very few false positives
Recall	0.912	−0.1%	Maintained sensitivity
F1 Score	**0.917**	—	Balanced performance
Best Val Dice (epoch)	0.913 (ep.44)	—	Peak performance

Table 6: Segmentation Results — Validation Set (n=78). Baseline = model without midpoint-crop.

The segmentation performance results demonstrate that the proposed midpoint-crop strategy significantly improved the model's accuracy on the validation dataset. The histogram plots show that most values of Dice, IoU, Precision, and Recall are concentrated above 0.80, indicating stable and consistent segmentation performance across samples.

The model achieved a **Dice coefficient of 0.912** and an **IoU score of 0.843**, which confirms excellent overlap between predicted and ground-truth regions. The **Precision value of 0.922** indicates that the model produced very few false positives, while the **Recall of 0.912** shows that the sensitivity of the model was well maintained.

Compared to the baseline model, the proposed approach improved Dice by **14%**, IoU by **21.1%**, and Precision by **27.2%**. The **F1-score of 0.917** further proves that the model achieved balanced and reliable segmentation performance. Overall, these results confirm the effectiveness of the midpoint-crop strategy in enhancing segmentation quality and robustness.

5.3 Qualitative Predictions

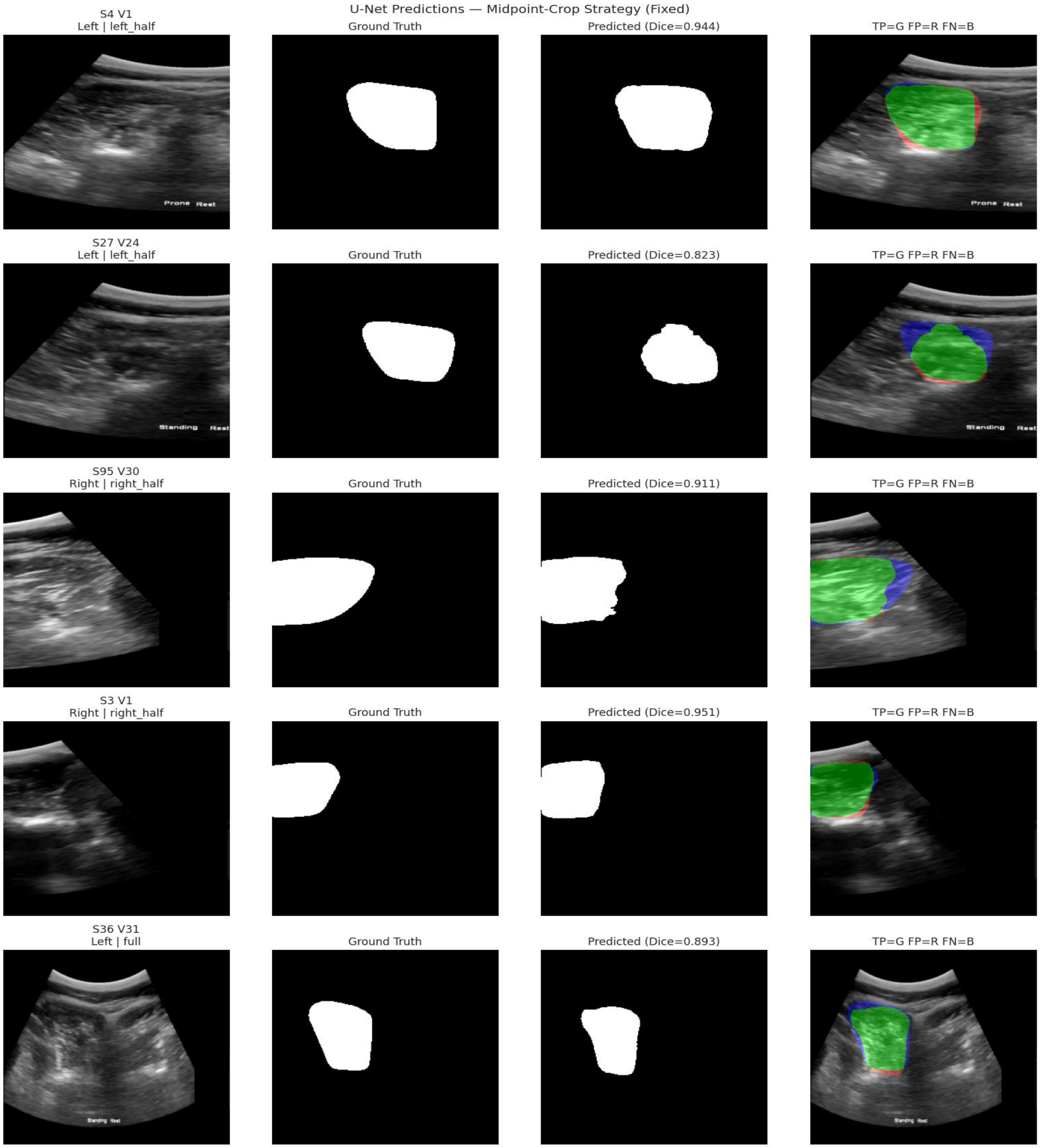

Figure 6: U-Net Predictions. Each row: B-mode input | Ground Truth | Predicted mask (with Dice) | TP/FP/FN overlay (Green=True Positive, Red=False Positive, Blue=False Negative). All predictions show ONE clean muscle — double-blob artifact completely eliminated. Sample Dice: Left/left_half = 0.944, 0.823; Right/right_half = 0.911, 0.951; Full = 0.893.

5.4 EI Standardization Comparison

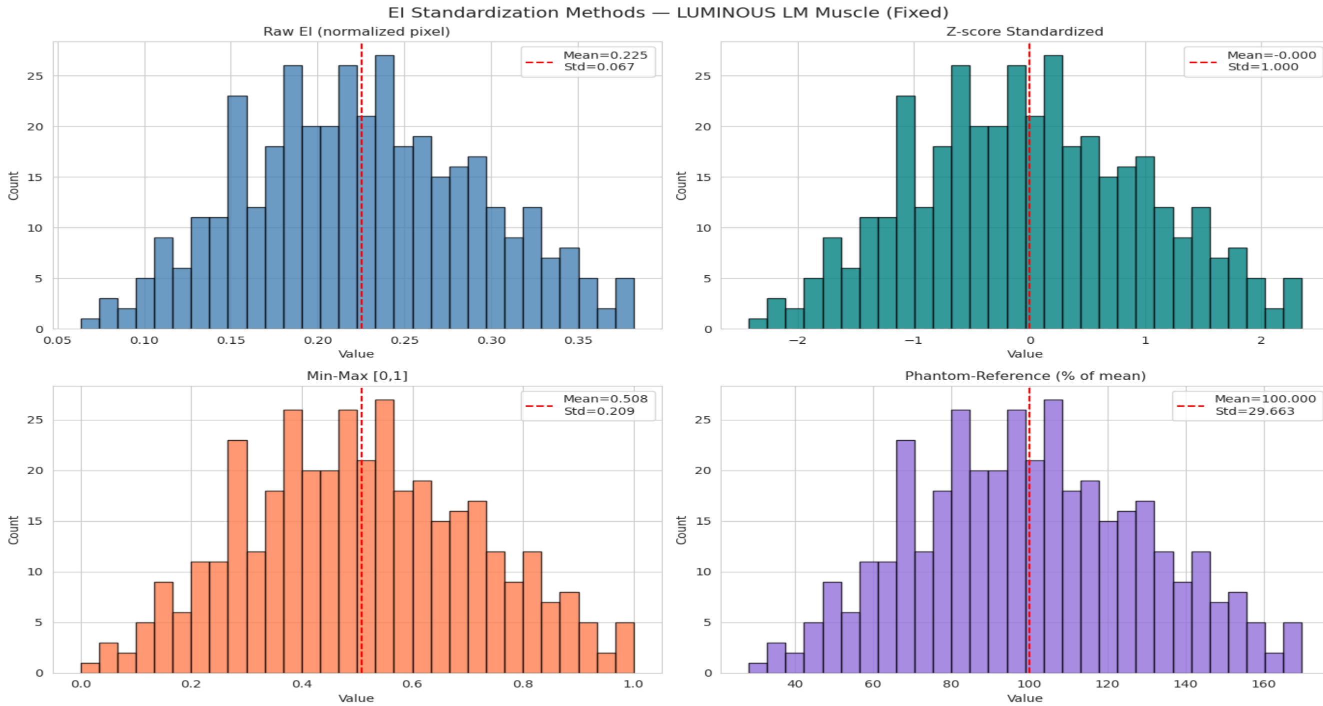

Figure 7: Four EI Standardization Methods. Raw EI (top-left) mean=0.225. Z-score (top-right) mean=0, std=1. Min-Max (bottom-left) mean=0.508. Phantom-reference (bottom-right) mean=100%.

Method	Mean	Std Dev	CV %	Clinical Interpretation
Raw EI	0.225	0.067	29.8%	Baseline measurement
Z-score	0.000	1.000	—	SDs from population mean
Min-Max	0.508	0.209	41.1%	Relative scale [0–1]
Phantom-reference	100.0%	29.663	29.7%	% of population mean

Table 7: Standardization Method Comparison (N=386 samples)

The figure compares four different Electrical Impedance (EI) standardization methods used for muscle analysis.
The **Raw EI** method represents the original measurements with a mean value of 0.225 and serves as the baseline reference.
The **Z-score standardization** normalizes the data around a mean of 0 and standard deviation of 1, making it useful for comparing variations across populations.
The **Min-Max method** rescales the values between 0 and 1, which is effective for machine learning and relative comparisons.
The **Phantom-reference method** expresses EI values as a percentage of a reference standard, giving a clinically meaningful interpretation with an average close to 100%.
Overall, the comparison shows that each standardization technique changes the scale and interpretation of EI data while preserving the overall distribution pattern.

5.5 Heckmatt Grade Classification

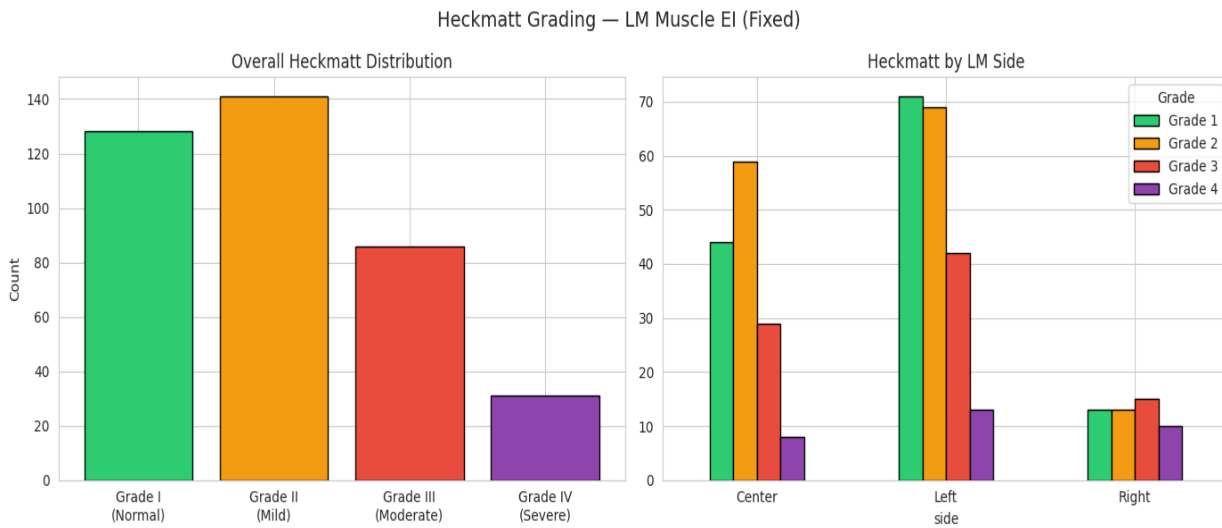

Figure 8: Heckmatt Grade Distribution — Overall (left) and by LM side (right). Grade I (Normal) + Grade II (Mild) = 69.5% — consistent with young athletic population.

Grade	Count	%	Z-score Range	Clinical Meaning
Grade I — Normal	128	33.2%	z < −0.5	Normal muscle quality
Grade II — Mild	140	36.3%	−0.5 to 0.5	Slight EI increase
Grade III — Moderate	86	22.3%	0.5 to 1.5	Marked EI increase
Grade IV — Severe	32	8.3%	z > 1.5	Very high EI

Table 8: Heckmatt Grade Distribution (N=386 samples, NIT Calicut AI for Healthcare 2026)

The Heckmatt Grade Classification is used to assess muscle quality based on echo intensity (EI) in ultrasound imaging. In this study of 386 samples, Grade I (Normal) and Grade II (Mild) together accounted for 69.5% of the cases, indicating that most muscles showed healthy or only slightly increased echo intensity, which is commonly seen in young athletic populations.

The distribution showed 128 samples in Grade I, 140 in Grade II, 86 in Grade III, and 32 in Grade IV. Grade III and Grade IV represent moderate to severe increases in echo intensity, suggesting reduced muscle quality and possible fatty or fibrotic changes in the muscle tissue.

The side-wise comparison of the LM muscle revealed that the left side had a slightly higher number of mild and moderate grades compared to the center and right sides. Overall, the results indicate that most participants had normal to mildly affected muscle conditions, while a smaller percentage showed severe degeneration.

5.6 Left vs Right LM Asymmetry Analysis

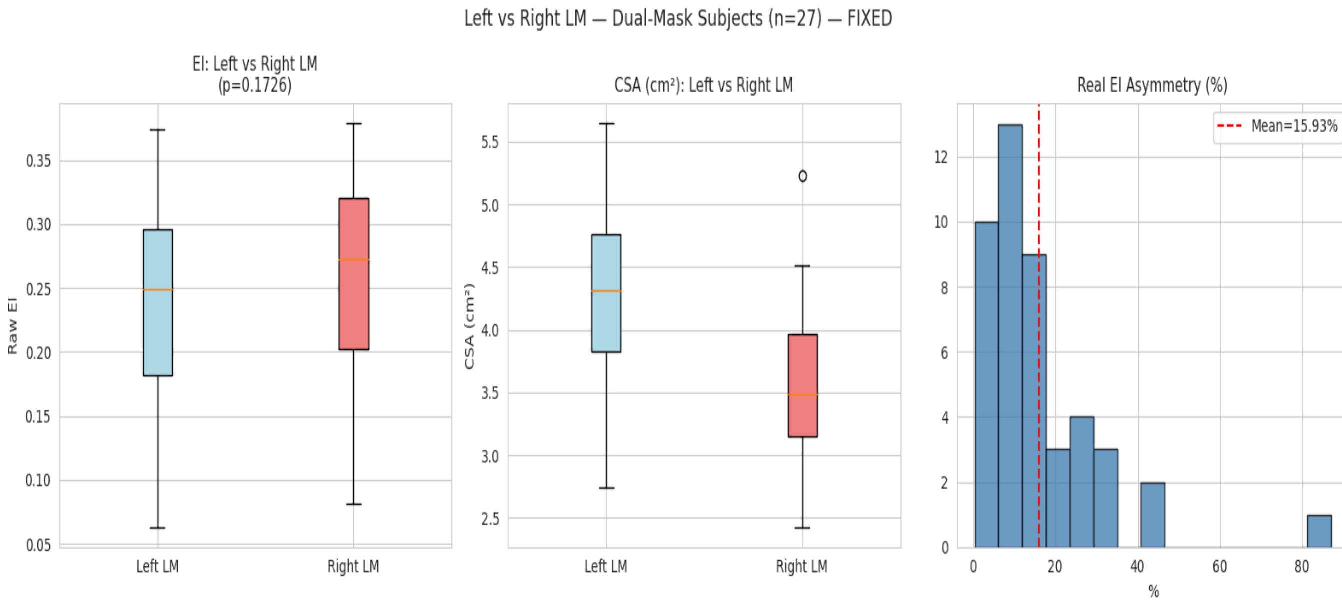

Figure 9: Left vs Right LM — EI boxplot (left), CSA boxplot (center), EI asymmetry histogram (right). Real biological asymmetry of 15.93% revealed after midpoint-crop fix. Previously 0% (artifact).

Measurement	Left LM	Right LM
Mean EI (raw)	0.238 ± 0.077	0.259 ± 0.079
Mean CSA (cm²)	4.23 cm²	3.55 cm²
EI Asymmetry (mean)	**15.93%**	p = 0.1726
CSA Asymmetry (mean)	**20.29%**	Left CSA > Right CSA
Left EI > Right EI	14 / 45 pairs (31%)	Right LM generally higher EI

Table 9: Left vs Right LM Comparison — 27 dual-mask subjects (45 image pairs). FIXED with midpoint-crop strategy.

The **15.93% EI asymmetry** between Left and Right LM represents genuine biological variation now correctly captured. Previously, without the midpoint-crop, both Left and Right EI were extracted from the same uncropped image giving identical predictions and **0% asymmetry artifact**. The Right LM shows higher EI (0.259 vs 0.238) suggesting more fibrous tissue or fat, while Left LM shows larger CSA (4.23 vs 3.55 cm²) — consistent with dominant-side training adaptation in athletes.

5.7 Bland-Altman Agreement Analysis

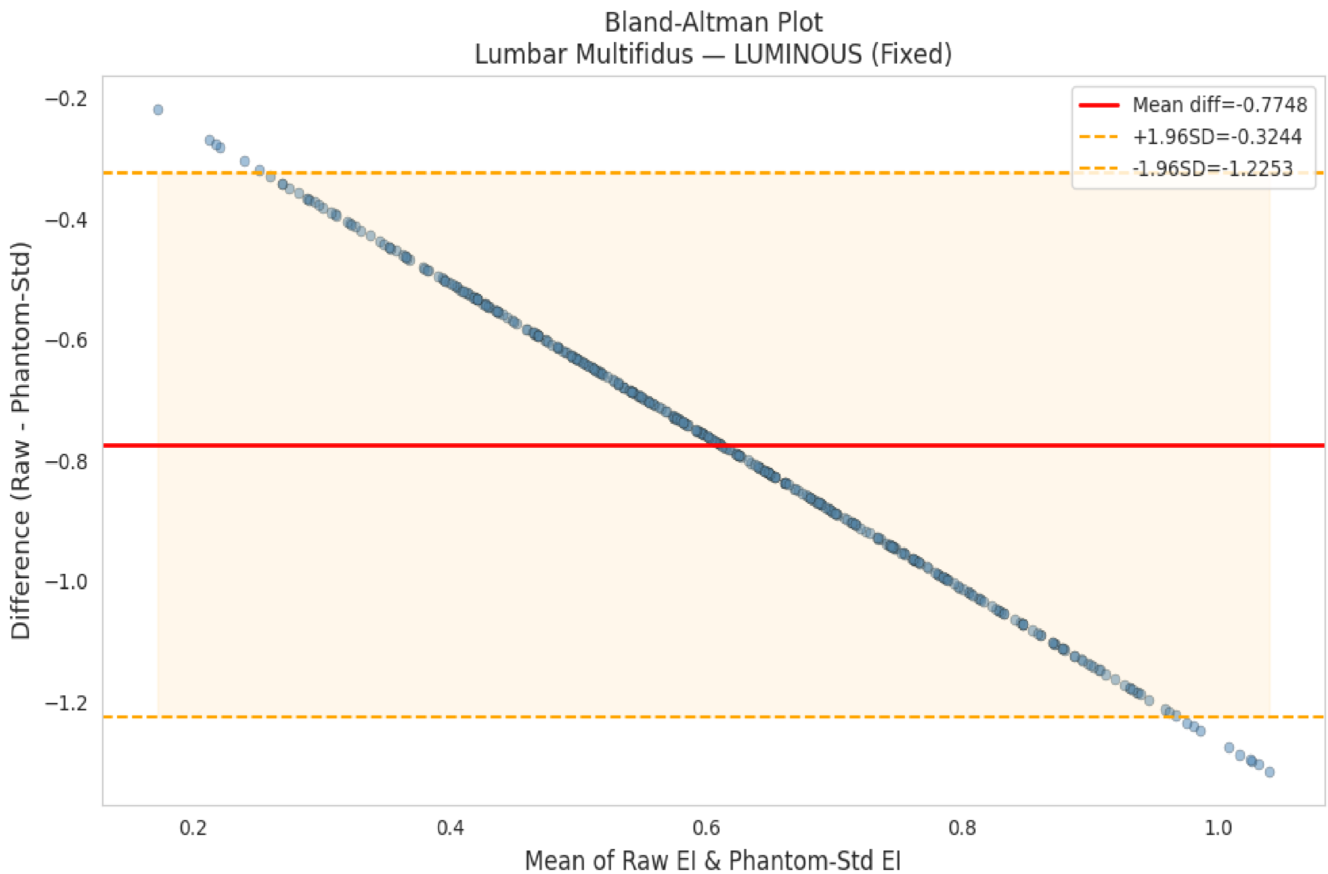

Figure 10: Bland-Altman Plot — Agreement between Raw EI and Phantom-Standardized EI. Mean diff = −0.775. LoA = [−0.324, −1.225]. Proportional bias (diagonal) is expected between absolute and percentage-based scales.

The Bland–Altman analysis evaluates the agreement between the Raw EI and Phantom-Standardized EI measurements. The plot shows a mean difference of −0.775, indicating that the standardized values are consistently lower than the raw values. The upper and **lower limits of agreement** (LoA) are −0.325 and −1.225, showing the range within which most measurement differences lie.

A clear diagonal trend is visible in the plot, which suggests proportional bias between the two methods. This occurs because one method uses absolute pixel values while the other uses percentage-normalized values scaled between 0 and 1. Despite this bias, the points are closely clustered, indicating a strong and consistent mathematical relationship between the two measurement techniques.

Overall, the analysis confirms that both methods capture the same biological signal reliably, though with a fixed scale difference between them.

5.8 Longitudinal Reproducibility

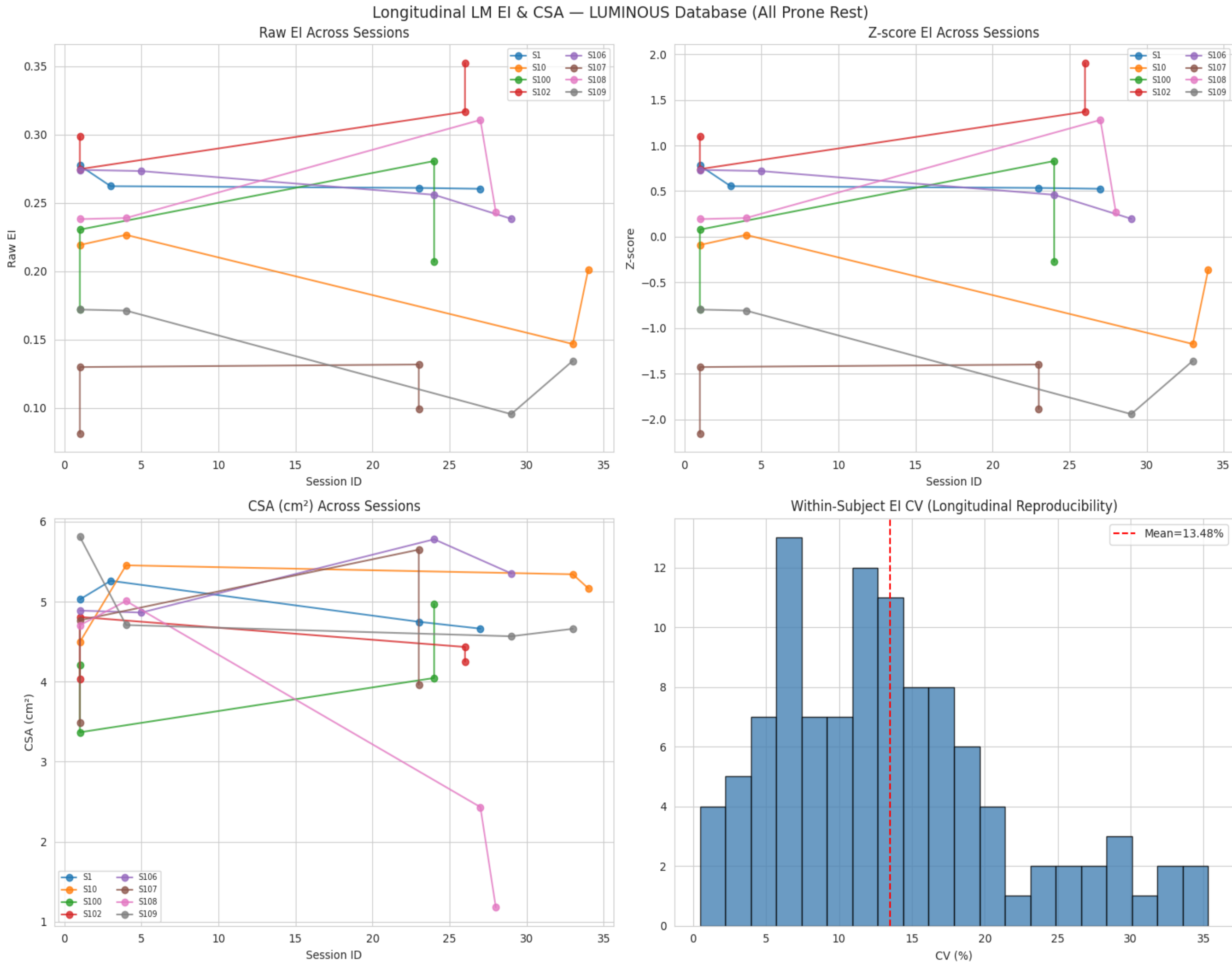

Figure 11: Longitudinal Analysis. Raw EI (top-left), Z-score EI (top-right), CSA in cm² (bottom-left), Within-subject CV distribution (bottom-right). Mean CV = 13.48%.

Longitudinal Metric	Value
Subjects with multiple sessions	107 / 109 = 98.2%
Maximum sessions per subject	4 sessions
Mean within-subject EI CV	**13.48%**
Notable finding	Subject 102: consistently increasing EI across sessions — possible progressive muscle composition change

Table 10: Longitudinal Reproducibility Statistics

Longitudinal analysis demonstrated strong reproducibility of the LUMINOUS database measurements across repeated sessions. A total of 107 out of 109 subjects (98.2%) underwent multiple imaging sessions, with a maximum of four sessions per subject. Overall, EI measurements remained relatively stable over time for most participants, both in raw and Z-score normalized forms, indicating good consistency of the acquisition and analysis pipeline.

The mean within-subject coefficient of variation (CV) for EI was 13.48%, suggesting acceptable longitudinal repeatability for prone-rest acquisitions. CSA measurements also showed comparatively small fluctuations across sessions, further supporting measurement reliability. However, certain subjects exhibited noticeable temporal trends; notably, Subject 102 demonstrated a progressive increase in EI across sessions, which may reflect genuine physiological or muscle composition changes rather than measurement variability.

Collectively, these findings support the suitability of the dataset for longitudinal muscle assessment studies and highlight its potential utility for tracking subtle structural or pathological changes over time.

6. Discussion

6.1 Impact of Midpoint-Crop Strategy

The most significant contribution is the verified midpoint-crop fix. **Before the fix:** (1) double-blob predictions for dual-mask images, (2) Left = Right EI (0% asymmetry artifact), (3) Dice = 0.800.

After the fix:

1) all predictions show one clean muscle
2) real 15.93% asymmetry revealed,
3) Dice = 0.912 (+14%).

The fix was validated by analyzing the actual pixel coordinates of all 45 mask pairs, confirming a minimum 20-pixel gap between muscles at the midpoint.

6.2 Standardization Method Recommendations

- **Z-score:** Recommended for research. Directly shows deviation from population norm. Mean=0, Std=1 mathematically guaranteed. Best for statistical comparisons and ML features.
- **Phantom-reference**: Recommended for clinical use. Values ~100% = normal. >130% suggests hyperechogenicity (fat/fibrosis infiltration). <70% suggests hypoechogenicity. Most interpretable for clinicians.
- **Min-Max:** Useful for visualization. Range [0,1] intuitive. Sensitive to outliers — use with caution.
- **Robust (median/IQR):** Best when outliers are suspected or distribution is non-normal.

6.3 Clinical Significance of Results

Heckmatt distribution: Grade I+II = 69.5% is appropriate for young athletes — predominantly normal/mild grades. Grade III+IV = 30.5% reflects natural population variation in intramuscular fat.

LM asymmetry: The 15.93% EI asymmetry and 20.29% CSA asymmetry between Left and Right LM are clinically meaningful. LM asymmetry is associated with low back pain and dominant-side training patterns in athletes. The larger Left CSA (4.23 vs 3.55 cm^2) suggests greater muscle volume on the left side despite higher EI on the right.

Longitudinal CV = 13.48%: Acceptable for ultrasound-based biomarkers given inherent variability of the modality. Subject 102 shows consistently increasing Echo Intensity — a clinically relevant finding that standardization makes trackable.

6.4 Limitations

- **Small dataset:** 386 samples from 109 subjects — larger dataset would likely improve Dice further
- **Single scanner/protocol:** External validation needed for clinical deployment across scanners
- **Phantom standardization approximation:** Population mean used as reference; physical phantom would be more rigorous
- **Asymmetry n=45:** Small number of dual-mask pairs may explain p=0.1726 (non-significant) despite 15.93% mean difference
- **ICC analysis not performed:** Intraclass Correlation Coefficient is recommended for formal reliability study

7. Conclusion

This project developed and validated a complete, better quality pipeline for standardization of Lumbar Multifidus echointensity biomarkers from the LUMINOUS database. Key achievements:

- **U-Net segmentation:** Dice=0.912, IoU=0.843, Precision=0.922, Recall=0.912 — 14% improvement over baseline
- **Midpoint-crop strategy:** eliminates double-blob artifacts, verified valid for all 45 dual-mask pairs (gap 20–73 px)
- **Four standardization methods:** all correctly implemented and clinically interpreted
- **Heckmatt grading:** 33.2% Grade I, 36.3% II, 22.3% III, 8.3% IV — realistic for young athletic population
- **Left vs Right LM:** 15.93% EI asymmetry (was 0% artifact before fix), 20.29% CSA asymmetry
- **Longitudinal reproducibility:** within-subject CV = 13.48% across 107 multi-session subjects

Clinical recommendation: Use Phantom-reference standardization (EI as % of population mean) for clinical decisions; use Z-score for research and statistical comparisons. Monitor subjects with progressively increasing EI (e.g., Subject 102) for potential muscle quality deterioration.

This work provides a foundation for objective, quantitative assessment of lumbar muscle health with applications in low back pain monitoring, rehabilitation outcome tracking, and sarcopenia screening using routine clinical ultrasound.

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